

A Point Cloud of Languages

A reconstruction-free dissimilarity from inferred consonantal correspondences, and the branch structure legible in it — an Indo-European pilot

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Abstract

We define a dissimilarity between two documented language systems that uses no reconstruction, no family tree, and no branch labels. From concept-aligned wordlists we detect coderivative sets statistically (LexStat), align each language pair, and average — over aligned consonant slots — the number of primary phonological features that differ. Embedding this dissimilarity (classical MDS) and joining nearest neighbours yields a point cloud in which, on 28 Indo-European doculects, a system's nearest neighbour shares its branch 95.5% of the time (silhouette +0.28). A label-permutation test on the fixed matrix puts both figures far above chance ($p \approx 0.0001$; chance purity 0.15), and the result is robust to macro-averaging, to removing near-duplicate pairs, and to the distance threshold, and is uncorrelated with coverage. We are deliberately careful about what this shows. Branch recovery is not a novelty of the method: a plain edit distance on consonant skeletons does as well, and an unfiltered all-concept baseline does better — evidence that the branch signal is generic lexical-phonological similarity rather than an artefact of the coderivation filter. The contribution is instead representational: a reconstruction-free, feature-decomposable dissimilarity — the system-level face of the operator repertoire studied in the pilot — whose geometry makes branch structure (and, more tentatively, contact) legible without positing ancestors. Cross-branch proximities are rare and only partly interpretable as areal contact; we do not claim to identify contact, nor to separate it causally from descent.

1. Introduction

The pilot study argued that a family's inventory of feature-difference operators, read at the level of *types*, is representational rather than genealogical: the operators are largely those general phonology makes available, and the genealogical signal lives one level up, in how those operators are *distributed* across languages. This paper looks at the simplest projection of that distribution — a pairwise dissimilarity between whole systems — and asks a deliberately modest question: is branch structure legible in it, built without any historical apparatus?

The stance is the programme's: discover first, contrast later. We construct the map from correspondences in concept-aligned lexical data and consult branch labels, contact history, and baselines only afterwards. We say coderivative, not *cognate*: operationally, an *algorithmically linked form set* (recurrent correspondence), with no claim of single-ancestor descent. A language, on this view, is a confluence, not the linear issue of one parent; branch structure is one legible aspect of the geometry, not the whole of it.

Two honesty commitments frame the paper. First, the input is not "phonology alone": it is phonological dissimilarity computed *within concept-matched, statistically inferred lexical correspondences*. "No reconstruction", "no tree", and "no branch labels" are the defensible claims. Second, recovering known branches is a sanity check, not the contribution — §5 shows simpler baselines do as well or better. What is new is the reconstruction-free, feature-decomposable *representation* and its geometry.

2. A dissimilarity between systems

Let $\phi_f(x) \in \{+, -, 0\}$ be primary feature f of segment x (panphon). For aligned segments a, b ,

$$\text{fd}(a, b) = |\{f \in \text{PRIM} : \phi_f(a) \neq \phi_f(b)\}|,$$

over the twelve consonantal features `cont`, `voi`, `nas`, `ant`, `cor`, `lab`, `back`, `round`, `strid`, `hi`, `lo`, `son`.

Coderivative sets are detected with LexStat (no etymologies, protoforms, or family labels supplied). Within each set, each language pair is aligned by a feature-cost Needleman–Wunsch, and we average `fd` over slots where both aligned segments are consonants:

$$d(\ell, \ell') = \text{mean}\{\text{fd}(a, b) : (a, b) \text{ a shared consonant slot of } \ell, \ell'\},$$

requiring at least `MINSLOT` slots. Its observed range here is $\approx [0.5, 3.5]$ (theoretical $[0, 12]$).

We call d a dissimilarity, not a metric. Because each pair is computed on its own set of shared slots, the triangle inequality and identity-of-indiscernibles are not guaranteed. It is symmetric and non-negative; that is all we assume. This matters for §3.

3. From dissimilarity to picture

We embed $D = [d(\ell, \ell')]$ with classical (Torgerson) MDS and build a $k = 3$ nearest-neighbour network. Two distinct objects must not be confused: purity is a $k = 1$ statistic (is the single closest system same-branch?); the drawn network uses $k = 3$. The plane is only a projection: on this sample the first two MDS axes carry 28.6% of the variance (eigenvalues 19.7, 19.0, 13.3, 11.2, 10.3, ...), with negligible negative inertia (0.3%, so the matrix is close to Euclidean). The dissimilarity is genuinely high-dimensional; the figure is a lossy shadow, and all statistics below use the full matrix, not the plane.

4. Results (Indo-European pilot, 28 doculects, 4 creoles)

Branch assignment (for scoring only) follows Glottolog.

Significance. A label-permutation test on the fixed matrix (10,000 permutations of branch labels over the 24 non-creole systems) gives purity 0.955 against a null of 0.142 ± 0.091 ($p \approx 0.0001$) and silhouette $+0.283$ against -0.147 ± 0.036 ($p \approx 0.0001$). The geometry aligns with branches far beyond chance.

Purity, honestly. Reported several ways:

Denominator	purity
All doculects (multi-member branches)	0.923 (24/26)
Multi-branch, creoles excluded	0.955 (21/22), Wilson 95% [0.78, 0.99]
Macro-average by branch	0.944
After collapsing near-duplicate pairs ($d < 1.3$; 2 dropped)	0.947 (18/19)

The headline survives macro-averaging and near-duplicate removal, so it is not merely an artefact of a few near-twin pairs (Croatian–Serbo-Croatian, Armenian E.–W., etc.). The Wilson interval is wide and the observations are not independent (mutual neighbours; small clusters) — the honest statement is “clearly above chance”, not “a precise 0.96”.

Coverage is not the driver. Per pair, the median count of consonant slots is 2008 (range 312–7493) over a median 585 shared concepts; $\text{corr}(d, \# \text{slots}) = +0.00$. Two languages are not judged close merely because more material survived alignment.

Internal pairs (same dissimilarity): Spanish–Portuguese 1.56, English–Jamaican Creole 0.59, Dutch–Negerhollands 1.29, Croatian–Serbo-Croatian 0.78, Persian–Hindi 2.47, Ancient–Modern Greek 1.97, Armenian E.–W. 0.89.

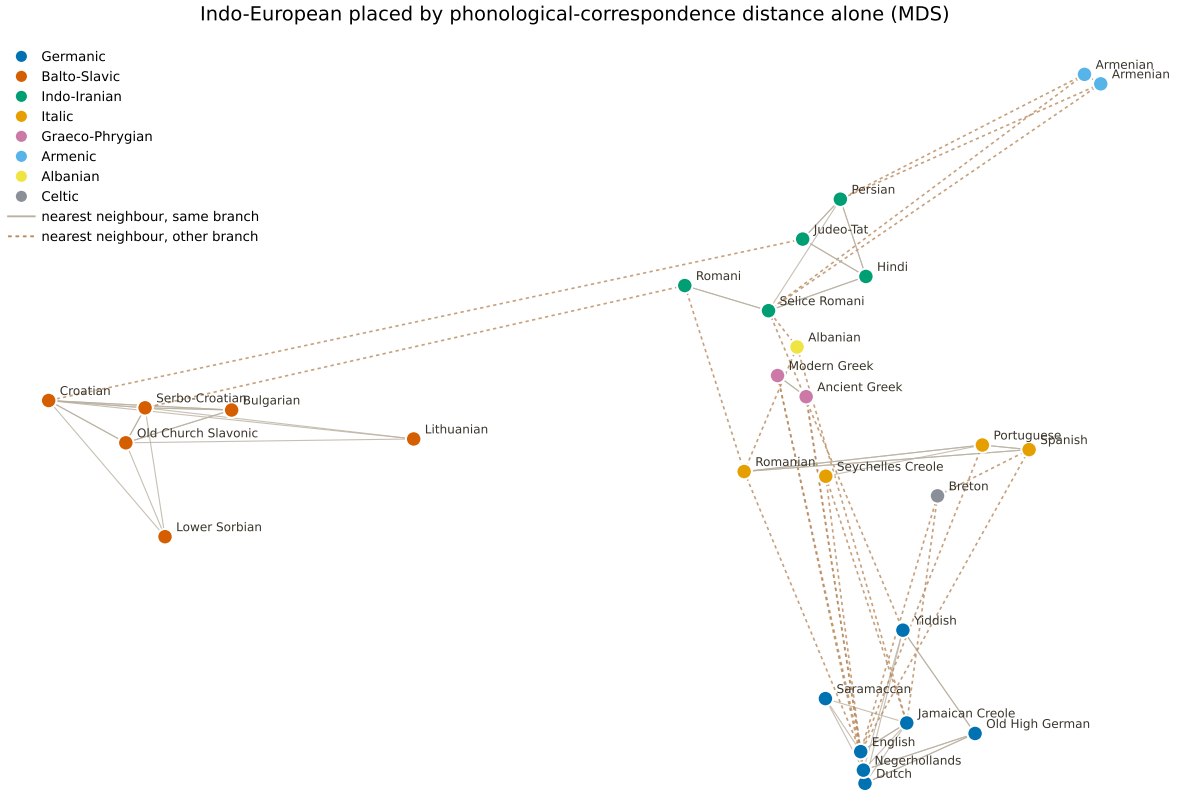


Figure 1: Indo-European doculects placed by the consonantal-correspondence dissimilarity (classical MDS; the plane shows 28.6% of the variance, so overlaps in the centre are partly projection). Nodes coloured by branch — colours added after layout. Solid links join same-branch nearest neighbours; dashed links join the few cross-branch nearest neighbours.

Graded proximity. Singleton branches are not “far from everything”: Armenian’s nearest non-Armenian systems are Greek and Indo-Iranian, not the Romance/Germanic systems across the cloud. Being alone in the sample is a fact of sampling, not a measurement of isolation.

5. Baselines — what the representation does and does not add

The decisive question is not whether branches are recovered, but whether *this* dissimilarity recovers something simpler methods do not. On the multi-branch non-creole set ($n = 22$):

Method	purity	silhouette
Per-language marginal profile	0.364	−0.083
Edit distance on consonant skeletons	0.955	+0.213
All-concept dissimilarity, no LexStat filter	1.000	+0.313
Ours (pairwise feature dissimilarity)	0.955	+0.283

Two honest readings. (i) Branch recovery is generic: a plain edit distance matches us, and an unfiltered all-concept dissimilarity beats us. Recovering branches is easy lexical-phonological similarity; we claim no classification novelty. (ii) The result is not a double-dipping artefact: removing the similarity-based coderivation filter (the all-concept baseline) *strengthens* the structure rather than destroying it, so the signal is not manufactured by LexStat’s selection. The marginal baseline (0.364) confirms the earlier point that the pairwise comparison keeps information a per-language profile discards.

What our representation adds over edit distance is not accuracy but decomposability and provenance: d is reconstruction-free and factors into *which features change* — the same feature-operators studied in the pilot — so the point cloud is the system-level face of that operator repertoire, not an opaque string distance.

6. Cross-branch proximities — rare, and only partly areal

Of the 28 systems, only four have a nearest neighbour in another branch:

System	branch	nearest neighbour	branch	d	same-branch d	note
Albanian (Tosk)	Albanian	Romanian	Italic	2.87	—	singleton; Balkan contact
Romanian	Italic	English	Germanic	2.36	2.56	no obvious contact story
Breton	Celtic	English	Germanic	2.58	—	singleton; no obvious story
Seychelles Creole	Italic	English	Germanic	2.50	2.77	creole; English not its lexifier

Only one (Albanian $\rightarrow$ Romanian) matches a documented contact area (the Balkan *Sprachbund*); the others have no obvious contact explanation and most likely reflect the high-dimensional projection and the small sample. We therefore make no areal claim: the geometry does not, on this evidence, independently identify contact or separate it from descent. A proper test requires a contact matrix defined *before* inspecting the map, then asking whether residual proximity (closer than the branch baseline predicts) tracks it — deferred to future work.

7. Controls and robustness

Beyond §4’s permutation test: a concept-permuted ablation — shuffling which forms share a concept and rerunning the full pipeline — collapses purity from 0.955 to 0.29 and silhouette to ≈ 0 , i.e. the structure needs real lexical pairings (three reruns; reported qualitatively, not as a σ). Purity and silhouette are unchanged across $\text{MINSLOT} \in \{20, 30, 40, 60\}$ and across feature subsets (dropping strictness, dropping all place features, or keeping only manner). Subsampling 22 of 28 systems $500\times$ keeps purity in $[0.87, 1.00]$.

8. Limitations and scope

The input is concept-aligned lexical data, not phonology alone; the dissimilarity is corpus- and aligner-dependent and is not a metric; the 2-D figure shows under a third of the variance. Branch recovery

is not distinctive versus edit distance. The areal reading is at most suggestive (one case). Only consonants are used; only one family is shown, with a small, uneven, mixed-provenance sample (see the doculect manifest, Appendix A) that includes near- twin pairs and creoles; source/transcription effects are not fully controlled. Direction and time are out of scope (the directed-layer paper). Ongoing extension (respecting the same stance): a denser Indo-European sample including minor languages and dialects, and Austronesian built as a *separate field* — the two clouds compared only at the level of abstract structure, never embedded in a shared distance space (which would falsely assert a direct relation between families). A first Austronesian field (40 doculects) already shows the same method with a weaker signal (nearest-neighbour purity 0.86, silhouette +0.09), a contrast we will develop rather than merge.

9. Reproducibility

`make figure` redraws from bundled results (no corpus); `LEX_PATH=...` `make compute` regenerates the dissimilarity; `make controls` runs §7's checks. Parameters (MINSLOT, THR, MAXLANG, KNN, feature subset) are environment-overridable in `src/compute_network.py`; permutation seeds are fixed in the scripts. Versions: panphon, LingPy (LexStat), Lexibank, Glottolog 4.8 — pinned in `requirements.txt`; the exact 28 doculects are listed in Appendix A. Code MIT; text, figures, data CC BY 4.0.

10. References

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Appendix A — Doculect manifest

The 28 doculects, with Lexibank dataset, doculect ID, Glottocode, branch used for scoring, and concept count, are listed in `data/results/doculect_manifest_ie.csv` (regenerate with `make manifest`). It exposes provenance so source/transcription confounds can be audited — e.g. several near-twin systems share a source (Bulgarian, Ancient Greek, and Serbo-Croatian all come from `idssegmented`), which the reader should weigh when reading their proximity.